

Burial state controls the impact force and travelling distance of a movable cubic object in granular flows

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ABSTRACT

Gravity-induced granular flows can bury and displace movable objects, thereby increasing the difficulty of post-hazard search and rescue. However, the fundamental mechanisms governing the impact response, downstream displacement, and burial of movable objects remain poorly understood. In this study, three-dimensional discrete element method (DEM) simulations are used to investigate the interaction between a granular flow and a movable cubic object. Analytical models are also proposed for predicting the travelling distance of the object. By varying the flow thickness h , normalized by the object height h_c , and the impact velocity v , we identify two distinct regimes governed by the burial state of the object. In the unburied regime ($h/h_c \leq 1$), the object remains exposed and is mainly transported by frontal impact and lateral flow deflection. In the buried regime ($h/h_c > 1$), the object is overtopped and progressively engulfed, leading to stronger coupling with the surrounding granular mass. Results reveal that the normalized impact force is predominantly controlled by the dimensionless flow discharge across both regimes. The travelling distance, however, exhibits clear regime dependence, following a power-law relationship with the dimensionless flow discharge in the unburied regime, while being better controlled by the Froude number in the buried regime. Analytical models developed for the two regimes successfully reproduce the computed travelling distances and provide a physically interpretable framework for estimating displaced-object locations.

1. Introduction

Gravity-induced granular flows, such as rock avalanches, debris avalanches, and snow avalanches, are common geophysical hazards characterised by rapid motion, long runout distances, and high destructive potential (Iverson, 1997; Johnson & Campbell, 2017). When these high-speed granular flows encounter objects such as houses, vehicles, or infrastructure components, they can bury and displace them tens to hundreds of meters downstream (Lagmay et al., 2008; Collins et al., 2021). This poses a significant challenge for post-disaster search and rescue, as the final resting position of an affected object may differ greatly from its original location. A representative example is the 2015 Guangming debris avalanche in Shenzhen, China, which destroyed and buried 33 buildings with some structures displaced horizontally by up to 150 m (Luo et al., 2019). Such extreme displacement underscores the critical need for locating buried or displaced objects entrained by granular flow hazards. However, this has been hindered by our limited understanding of the interaction between a granular flow and a movable object that frequently involves complex multi-body, multi-scale and

multi-way interactions during a typical impacting process.

To date, numerous field, experimental, and numerical studies have quantified the dynamic impact of granular flows on fixed objects, such as rigid barriers and walls, under a wide range of conditions, including variations in grain-size distribution, material composition, and boundary constraints (Faug, 2015; Goodwin et al., 2021; Huang et al., 2022; Nagl et al., 2024). These studies have substantially advanced our understanding of flow-object interaction. However, their findings cannot be directly extended to movable objects. Unlike fixed ones, movable objects are governed by strongly coupled interactions among the granular flow, the object itself, and the basal surface, which makes their transport and final deposition positions particularly difficult to predict. This complexity is further amplified when the object is progressively buried by the incoming granular material during transport. In such cases, the object evolves from an initially exposed state to a partially or even fully buried state. This transition continuously alters the contact conditions, force balance, and effective mobility of the object (Fang et al., 2026), which in turn influences its subsequent transport trajectory and eventual resting location.

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Knowledge from related research fields provides a useful reference for understanding object mobility in granular flows. For example, in coastal engineering, the transport of individual blocks by tsunami waves has been extensively investigated, revealing the important roles of object shape (Goto et al., 2007) and density (Nandasena and Tanaka, 2013) in controlling object mobility. Fortunately, in recent years, the attention has begun to shift from fixed obstacles to movable objects in granular flow. Feng & Song (2025) showed that the deposition position of the object is predominantly controlled by the basal sliding velocity of incoming granular flow. Fang et al. (2026) developed a novel coupled numerical framework for simulating non-spherical objects entrained in flow-type landslides, providing a numerical basis for studying movable-object dynamics in granular flows. Nevertheless, the governing mechanism remains poorly understood. In particular, it is still unclear how the burial state alters the impact response and transport behaviour of a movable object, whether the same dimensionless parameters govern impact force and travelling distance, and how these processes can be predicted within a unified physical framework.

Physical experiments or numerical simulations provide powerful tools for elucidating the underlying interaction mechanisms between a flow and a movable object, enabling more reproducible and detailed analysis under well-controlled conditions. However, in dense and highly mobile granular flows, the internal motion of an entrained object is difficult to observe directly because of the strong opacity of the surrounding material (Sun et al., 2024; Escobar et al., 2025). As a result, key processes such as impact, burial, and subsequent transport are difficult to resolve experimentally in sufficient detail. The discrete element method (DEM) therefore provides an essential alternative. DEM enables direct tracking of individual particles and provides real-time information on particle contacts and positions throughout the simulation (Wang et al., 2025), thereby providing a clear basis for monitoring the response and movement of a buried object. Furthermore, DEM provides a range of approaches for representing non-spherical and irregular particles or objects, including superquadrics, polyhedral or spheropolyhedral particles, level-set or mesh-based descriptions, and multisphere clumps (Zhao et al., 2023). Owing to these capabilities, DEM is adopted here for studying the complex interactions between granular flow and movable object.

In this study, we conduct a series of three-dimensional DEM simulations to investigate the impact, burial, and transport of a movable object subjected to granular flows. The object is idealized as a cube to facilitate a systematic analysis of its motion and interaction with the surrounding granular material. Unlike fixed-obstacle impact problems, in which the obstacle remains stationary and the main concern is the impact force acting on a fixed structure, the present study treats the cube motion, burial state, impact response, and travelling distance as coupled outcomes of granular-flow impact. In contrast to previous studies on tsunami boulders and movable objects, this work places particular emphasis on the role of burial, which can continuously modify the flow-object interaction mechanism, alter the force balance acting on the object, and ultimately affect its downstream displacement. Specifically, we examine how burial state influences the impact force, object mobility, and final travelling distance. Based on the simulation results, new analytical models are further developed to predict the travelling distances of both unburied and buried objects within a unified physical framework. The main objective of this study is to improve the fundamental understanding of granular flow-object interactions and to provide useful insights for locating buried or displaced objects in post-disaster search and rescue operations.

2. Theoretical framework

2.1. Unburied object dynamics

This section develops a simple analytical model for the transport of an unburied cubic object by a granular flow, which is treated as a two-

dimensional dynamic problem, as illustrated in Fig. 1a. The incoming flow is assumed to be continuous, steady, and uniform, with depth-averaged velocity v , height h , length L and bulk density ρ . When impinging on the object, the conservation of momentum for both the cubic object and the granular flow travelling at the same velocity v_0 can be expressed as follows:

$$\rho L h v = (\rho L h + \rho_c h_c^2) v_0 \quad (1)$$

where ρ_c and h_c are the density and the height of the cubic object, respectively. Based on the assumption that the kinetic energy is transformed into frictional dissipation energy without any other significant loss of energy, one can consider the total energy of the cube and the granular flow after travelling distance up to s :

$$E(s) = \frac{1}{2}(\rho L h + \rho_c h_c^2) v_0^2 - (\mu \rho L h g + \mu_c \rho_c h_c^2 g) s, \quad (2)$$

where g is the gravity, μ is the friction coefficient between flow and flume bed, μ_c is the friction coefficient between object and flume bed. By setting $E(s) = 0$, combining Eqs. 1 and 2 yields the final travelling distance of the object, as follows:

$$s = \frac{(\rho L h v)^2}{2g(\rho L h + \rho_c h_c^2)(\mu \rho L h + \mu_c \rho_c h_c^2)} \quad (3)$$

By assuming $\mu = \mu_c$, Eq. (3) can be written in the following form:

$$s = \frac{(\rho L h v)^2}{2\mu g(\rho L h + \rho_c h_c^2)^2} \quad (4)$$

2.2. Buried object dynamics

Fig. 1b illustrates the cubic object transported and buried by the incoming flow. The analytical model presented in this section is based on the assumption that the motion of the object can be described by Newton's second law, as follows:

$$\rho_c h_c^2 \frac{dv_0}{dt} = \alpha \rho h_c (v - v_0)^2 + F_R \quad (5)$$

where α is the dynamic pressure coefficient. The first term on the right-hand side is the dynamic impact force of incoming flow against the object. The second term represents the resultant of the remaining force acting on the object. A detailed expression for F_R is

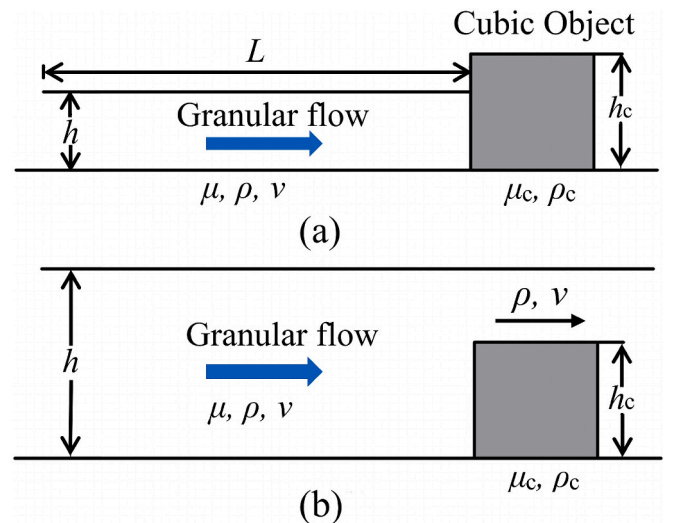


Fig. 1. Sketches of granular flow impacting the movable cubic object: (a) unburied regime and (b) buried regime.

$$F_R = \mu_1 \rho (h - h_c) h_c g + \int_{h-h_c}^h k \rho g z dz - \mu_c g [\rho (h - h_c) h_c + \rho_c h_c^2] \quad (6)$$

where $\mu_1 \rho (h - h_c) h_c g$ is the friction force on the object by overlying flow, μ_1 is the friction coefficient between flow and object, $\mu_c [\rho (h - h_c) h_c g + \rho_c h_c^2]$ is the basal friction of the object, $\int_{h-h_c}^h k \rho g z dz$ is the lateral pressure forces. k is the earth pressure coefficient. Previous work commonly assumed $k = 1$ for sheared granular flows in the steady and uniform regime (Silbert et al., 2001). Subsequent investigations revealed a slight normal-stress anisotropy even in this regime with $k = 1.05$ (Weinhart et al., 2016). It is likely that k may deviate substantially from unity and can be either smaller than 1 or greater than 1 under active or passive states (Faug, 2020). By assuming $w = v - v_0$, followed by some algebraic rearrangement, Eq. (5) can be recast into

$$\frac{dw}{dt} = -Gw^2 - J \quad (7)$$

where

$$G = \frac{\alpha \rho}{\rho_c h_c} \quad (8)$$

and

$$J = \frac{F_R}{\rho_c h_c^2} \quad (9)$$

are constant coefficients. Here, J represents the acceleration contribution associated with the remaining resultant force F_R . It should be noted that F_R includes both driving and resisting components. Therefore, J is not always positive. When $J < 0$, the F_R acts as a net resistance. In this case, complete synchronization with the flow is not physically sustainable, because at $w = 0$ the dynamic impact term vanishes whereas the net resisting force remains. When $J = 0$, the object is accelerated only by the dynamic impact term, which is proportional to the square of the relative velocity w^2 . As the object velocity approaches the flow velocity, w decreases and the dynamic impact force progressively vanishes. Therefore, the object velocity approaches the flow velocity only asymptotically. The following solution of Eq. (7) is therefore valid only for the admissible condition $J > 0$, or equivalently $F_R > 0$, as follows:

$$w(t) = q \tan(-Gqt + C) \quad (10)$$

where $q = \sqrt{J/G}$ and C is a constant of integration. The value of C is constrained by the fact that $w = v$ when $t = 0$, which yields

$$w(t) = q \tan[-Gqt + \phi] \quad (11)$$

where $\phi = \arctan(v/q)$. Then the object velocity at time t can be written in the explicit form

$$v_0(t) = -q \tan[-Gqt + \phi] + v \quad (12)$$

After t_a , the object's velocity is equal to the velocity of granular flow. t_a is given as

$$t_a = \phi / (Gq) \quad (13)$$

The travelling distance of the object driven by incoming flow is determined by integrating Eq. (12) from $t = 0$ to t_a to find that

$$s_1 = \int_0^{t_a} v_0(t) dt = \frac{v\phi}{Gq} + \frac{\ln(\cos\phi)}{G} \quad (14)$$

Subsequently, we assumed that both the cube and the granular flow travel at the same velocity. Conservation of total energy tells us that kinetic energy will be dissipated by basal friction, neglecting flow-object interactions, which yields

$$s_2 = \frac{v^2}{2\mu g} \quad (15)$$

Consequently, the total travelling distance s can be obtained by combining Eq. (14) and Eq. (15), as follows:

$$s = s_1 + s_2 \quad (16)$$

To further evaluate the influence of empirical parameters in the buried-regime analytical model, a parameter-sensitivity analysis is conducted for α , μ_c , μ_1 , and k (Fig. 2). The travelling distance s is calculated using Eq. (16) by varying one parameter at a time while keeping the remaining parameters fixed at their reference values. Three representative burial ratios, $h/h_c = 2.0$, 4.0 , and 6.0 , are considered to examine the influence of burial depth.

The predicted travelling distance varies systematically with the four analytical-model parameters. As shown in Fig. 2a, increasing α leads to a gradual decrease in the predicted travelling distance for all three h/h_c , and the curves tend to approach an approximately stable value when α becomes large. A similar decreasing trend is observed for the earth pressure coefficient k (Fig. 2b), where s decreases rapidly at relatively small k and then gradually levels off. For the object-bed friction coefficient μ_c (Fig. 2c), the predicted travelling distance generally decreases as μ_c increases, reflecting the increased basal resistance acting on the cube. The effect of the flow-object friction coefficient μ_1 is shown in Fig. 2d, where s decreases more gradually with increasing μ_1 . These trends would not be changed in spite of different h/h_c . Overall, this parameter analysis illustrates how the simplified buried-regime analytical model responds to variations in the dynamic impact coefficient, lateral pressure coefficient, basal resistance, and flow-object friction.

3. Methods

3.1. Dimensionless numbers

To enable comparison across diverse geological and topographical settings, physical parameters are typically normalized into dimensionless forms. Such dimensionless frameworks provide a robust quantification of bulk flow characteristics and the relative dominance of mechanical stresses across different geophysical phenomena (Lanzoni et al., 2017; Du et al., 2023). In this study, we use dimensionless discharge (q^*) and Froude number (Fr), both of which are well-defined and have been widely used in previous studies (Iverson, 1997; Du et al., 2023). Fr characterizes the inertial state of the flow relative to gravity, whereas q^* reflects the amount of moving granular material and the associated momentum supply.

The dimensionless flow discharge (q^*) is defined by the flow discharge per unit width relative to the grain size (Parker et al., 2007):

$$q^* = \frac{vh}{\sqrt{gD^3}} \quad (17)$$

where D is represented by the mean grain size in this study. The Froude number is defined as the ratio of inertial to gravity force (Wenner et al., 2021):

$$Fr = \frac{v}{\sqrt{gh}} \quad (18)$$

The Froude number is commonly used to characterize free-surface flow regimes, including subcritical ($Fr < 1$), critical ($Fr = 1$), and supercritical conditions ($Fr > 1$).

3.2. Numerical model verification and parameter determination

The open-source DEM code LIGGGHTS is used here to simulate the impact of granular flow on the cubic object and thus determine the input

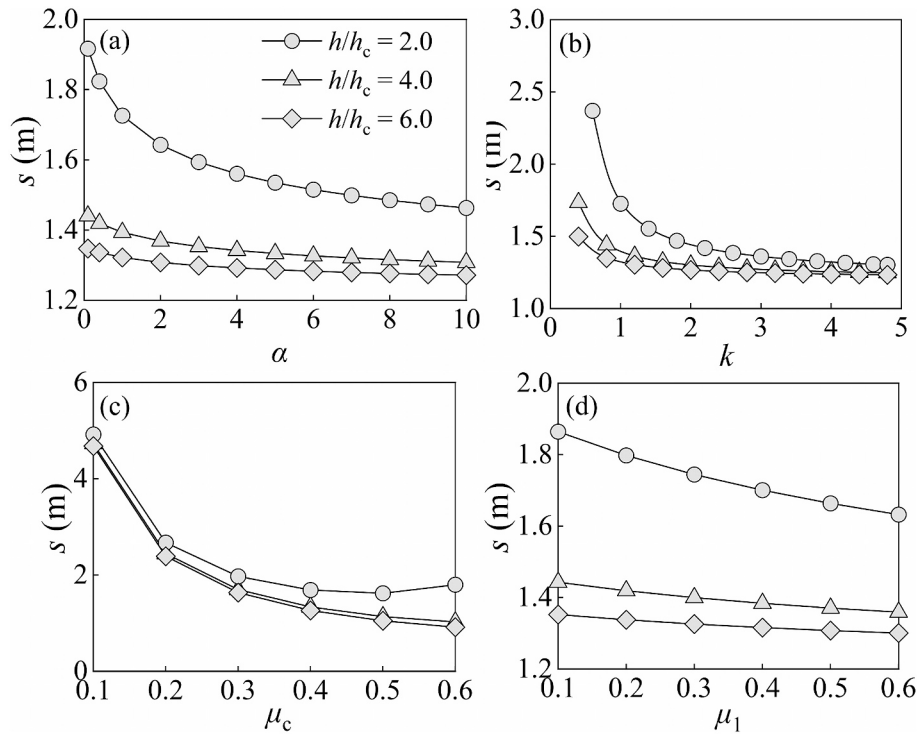


Fig. 2. Sensitivity of the predicted travelling distance s in the buried regime to the analytical model parameters: (a) dynamic pressure coefficient α , (b) earth pressure coefficient k , (c) object-bed friction coefficient μ_c , and (d) flow-object friction coefficient μ_1 . The travelling distance is calculated by varying one parameter at a time while keeping the other parameters fixed. The reference values are $\rho_c = 2550 \text{ kg/m}^3$, $h_c = 0.06 \text{ m}$, $v = 3 \text{ m/s}$, $L = 1.4 \text{ m}$, $\rho = 1755 \text{ kg/m}^3$, $\mu_c = 0.38$, $\mu_1 = 0.34$, $\alpha = 1.0$ and $k = 1.0$.

mechanical parameters. Fig. 3a illustrates the experimental setup composed of an inclined flume and a deposition area. The material storage container at the upstream end of the flume has a length of 0.4 m, height of 0.3 m, and width of 0.2 m. The spreading length of the material is 2.6 m and inclined to 20° . A flat plane measuring 1.8 m in length and 1.2 m in width is placed at the downstream end of the flume and serves as the deposition area to allow the full deposition of granular flow. A cubic object is placed along the centerline of the plane, which is positioned at 0.2 m away from the downstream end of the flume. The cubic object has a side length of 0.06 m and a density of 2550 kg/m^3 . Glass beads with an average diameter of 10 mm and a density of 2550 kg/m^3 are prepared into storage container to simulate the incoming flow. A total mass of 20 kg of glass beads is used.

To reproduce the experimental setup, all geometries simulated in the DEM are kept consistent. The glass beads are modeled using DEM particles with identical diameter and density. Although superquadric particles are available in the open-source version of LIGGGHTS, the cubic

object is represented here using the clump method. This method was adopted because the present study focuses on the burial-controlled impact and transport of a movable cubic object rather than on comparing different DEM shape-representation schemes. The clump method allows the cube to be constructed as a rigid assembly of spherical sub-particles, so that particle-object contacts can be resolved within the same sphere-based contact framework as that used for the glass beads. This provides a robust and straightforward treatment for dense granular-flow impact and burial processes involving numerous simultaneous contacts. In DEM, a Monte Carlo resampling procedure is used to assess properties that are needed for the rigid motion update. The spacing parameter SD is defined as $SD = l_s/l_d$, where l_d denotes the particle diameter and l_s denotes the centroid distance between the adjacent sub-particles, as illustrated in Fig. 3b. Thus, SD is dimensionless and controls the overlap between adjacent sub-particles. Since the sub-particle density is kept identical to that of the real cube, SD is calibrated to match the geometry and mass of the constructed clump with those of

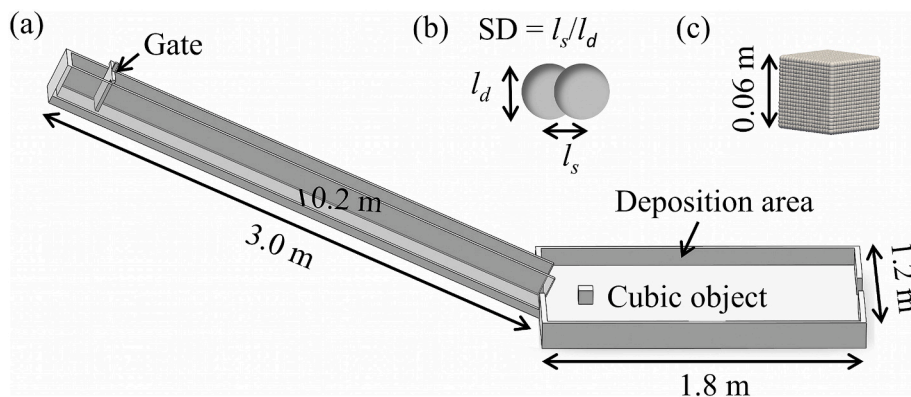


Fig. 3. (a) Flume experiment setup for DEM verification. (b) Definition of the dimensionless spacing parameter SD . (c) Illustration of cubic object simulated in DEM.

the real object. This gives $SD = 0.5$. In this study, the cubic object is constructed using 5321 sub-particles each with a diameter of 0.006 m (Fig. 3c). Table 1 summarizes the input parameters.

The motion of the movable cube in the flume experiment is recorded using a video camera during the impact and deposition process. The grids marked on the deposition plate, with a spacing of 10×10 cm, are used as spatial references for image calibration. The instantaneous position of the cube is identified from the recorded image sequence, and the travelling distance is measured as the horizontal displacement of the cube from its initial position. The measured and computed evolutions of the travelling distance (Fig. 4a) and the streamwise translational velocity of the cube (Fig. 4b) are presented. It is evident that the travelling distance predicted by DEM agrees well with the measured value of 0.68 m, and the velocity also agrees well with the experimental results with a difference in peak velocity of about 11%. For further quantitative comparisons, the snapshots of the experiment and simulation at three typical instants are also compared (Insets in Fig. 4a). When the flow impacts the object, the flow is deflected symmetrically around the block, forming a bow shock. The object interrupts the upstream material supply and lateral spreading becomes evident. Owing to the strong impact of incoming flow, the object is significantly displaced. The same sequence of flow deflection, bow-shock formation, lateral spreading, and object displacement is reproduced by the DEM simulation. The snapshots also show that the cube remains exposed and unburied in both the experiment and the numerical simulation, indicating that the DEM model captures the observed unburied transport state in this benchmark case. Nevertheless, this comparison should not be regarded as a comprehensive validation of all dynamic quantities. For example, direct measurement of the impact-force history is not provided here.

It should be noted that the apparent contact roughness of a multi-sphere clump depends on the sub-particle size. In the present study, the effective influence of clump surface roughness and contact behaviour is incorporated through calibration of the DEM friction parameters against the flume experiment. As shown in Fig. 4, the adopted clump representation, together with the calibrated contact parameters, reproduces the measured travelling distance, streamwise velocity, and the observed motion process of the cube reasonably well. Since the main quantities examined in this study are the bulk impact response and travelling distance of the cube, this agreement supports the adequacy of the adopted clump discretization for the present purpose.

3.3. Numerical model setup and simulation cases

Fig. 5 illustrates the model configuration for further systematic simulations of object transport and burial. A granular mixture sample, with a length of 1.4 m, a width of 0.3 m, and a prescribed initial height h , composed of tridisperse particles, is initially placed on the top of the flume channel behind a gate wall. The particles are uniformly assigned prescribed initial velocities v , and released to impact a cubic object with the height h_c of 0.06 m, consistent with the object shown in Fig. 3. By prescribing the initial velocity and flow thickness directly, we can control the bulk incoming conditions at the object location and systematically examine their effects on impact force, burial state, and

Table 1

Input parameters for the DEM numerical simulations.

Parameters	Values
Flume Young's modulus (Pa)	3.2×10^9
Particle Young's modulus (Pa)	6×10^{10}
Coefficient of restitution (dimensionless)	0.78
Particle density (kg/m^3)	2550
Gravitational acceleration (m/s^2)	9.81
Inter-particle friction coefficient (dimensionless)	0.34
Particle-flume friction coefficient (dimensionless)	0.38
Particle Poisson ratio (dimensionless)	0.25
Flume Poisson ratio (dimensionless)	0.35

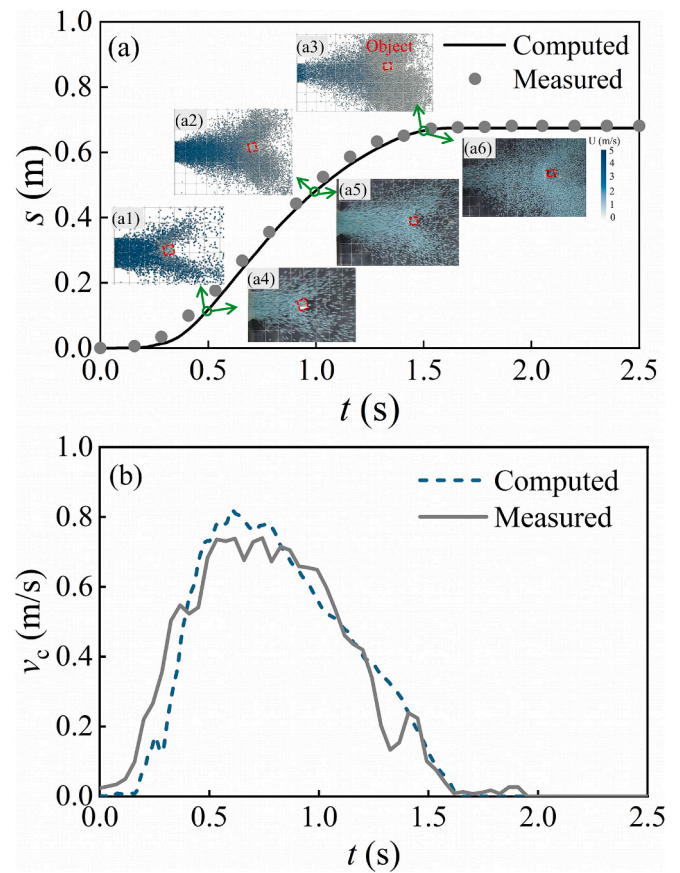


Fig. 4. (a) Comparison between measured and computed travelling distance of the object. Insets (a1-a3): computed flow kinematics at different instants. Insets (a4-a6): measured flow kinematics at different instants. (a1, a4): $t = 0.5$ s; (a2, a5): $t = 1.0$ s; (a3, a6): $t = 1.5$ s. The time instants are denoted by green open circles. (b) Comparison between measured and computed streamwise translational velocity of the cube.

travelling distance. A similar model setup has been adopted in previous granular/debris-flow impact simulations, where the propagation phase was disregarded to obtain well-defined impact conditions (Calvetti et al., 2019; Li et al., 2020).

The flume is sufficiently long to prevent the object from being driven out of the computational domain by the incoming flow. The mass percentages of the particles with diameters of 0.01 m, 0.014 m and 0.018 m are 20%, 20% and 60%, respectively. This simplified tridisperse particle-size distribution is not intended to reproduce a realistic natural grain-size distribution, such as a power-law or Gaussian distribution. Instead, it is adopted to introduce a controlled degree of polydispersity, which helps reduce artificial ordering or crystallization effects that may arise in monodisperse DEM assemblies, while avoiding the additional complexity associated with broad grain-size distributions, such as segregation and size-dependent mobility. We vary the initial height of granular sample h and the prescribed initial velocities v to obtain a broad range of Froude numbers (Table 2). The simulated Fr range broadly overlaps with the reported range for natural granular flows, ranging from 0.5 to 7.6. Varying h also changes the relative flow thickness, providing a simple indicator to distinguish shallow-flow cases ($h/h_c \leq 1$) from deep-flow cases ($h/h_c > 1$). The corresponding dominant burial state of the cube is then confirmed directly from the DEM simulations. In total, 56 simulations are performed. A fixed pseudo-random seed and the same particle-generation procedure are used to ensure repeatability of each numerical realization and to avoid introducing uncontrolled variations in the initial packing when comparing different test cases. This fixed-seed approach is not intended to assess robustness to different

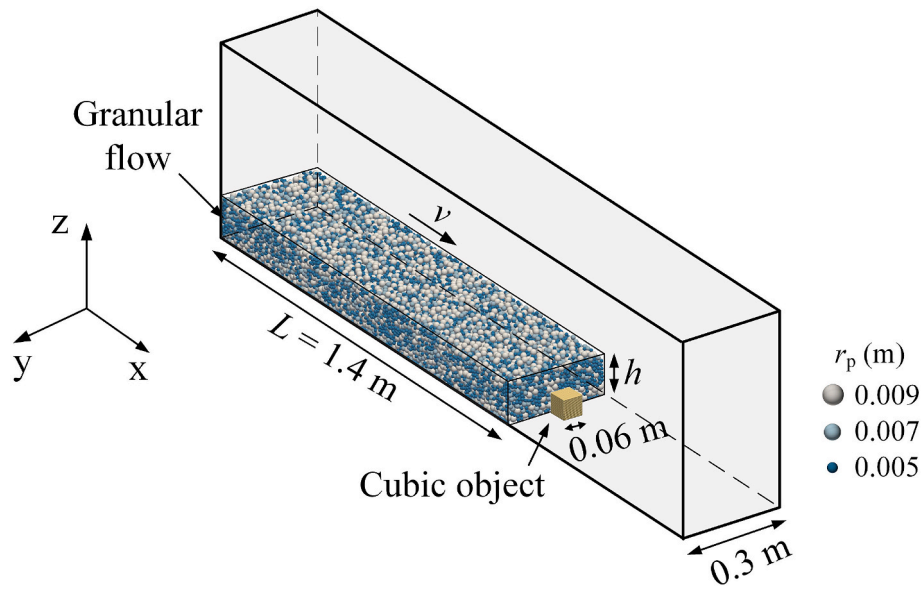


Fig. 5. Numerical model setup for granular flow impacting the cubic object.

Table 2
Summary of simulation program.

h/h_c	0.4	0.5	0.6	0.8	1	2	3	4	5	6
ν (m/s)	1.0, 1.2, 1.4, 1.6, 1.8	1.0, 1.2, 1.4, 1.6, 1.8	1.0, 1.2, 1.4, 1.6, 1.8	1.0, 1.2, 1.4, 1.6, 1.8	1, 2, 3	0.2, 1, 2, 3, 4, 5, 6	0.2, 1, 2, 3, 4, 5, 6	0.2, 1, 2, 3, 4, 5, 6	0.2, 1, 2, 3, 4, 5, 6	0.2, 1, 2, 3, 4
q^*	3.93 ~ 7.08	4.92 ~ 8.84	5.90 ~ 10.62	7.86 ~ 14.16	9.83 ~ 29.50	3.93 ~ 118	5.90 ~ 177	7.87 ~ 236	9.83 ~ 295	11.80 ~ 236
Fr	2.06 ~ 3.71	1.84 ~ 3.32	1.68 ~ 3.03	1.46 ~ 2.62	0.26 ~ 2.61	0.18 ~ 5.53	0.15 ~ 4.52	0.13 ~ 3.50	0.12 ~ 3.50	0.11 ~ 2.13

random initial packings. In each case, particles are initially generated at random positions in the simulation domain while ensuring no initial overlaps. The total number of generated particles ranges from 10,590 to 73462, depending on the initial height of granular sample.

In the DEM modelling, the walls (in the y -direction and negative x -direction) and the bed (negative z -direction) of the channel are treated as fixed, rigid, and frictional walls. The remaining boundaries are defined as open, allowing particles to exit from the simulation domain once they move beyond these limits. Contact forces between particles, as well as between particle and wall, are calculated using the Hertz contact law. Particle trajectories are obtained by numerically integrating the force and torque balance equations for each particle at a fixed time step $\Delta t = 1 \times 10^{-6}$ s, which satisfies the Rayleigh time and Hertz time criteria (Smuts, 2015). The input parameters are consistent with those listed in Table 1.

4. Results

4.1. Flow-object interactions

Fig. 6 illustrates the detailed impact process of granular flow on a movable cubic object and the particle velocity profiles along the flow thickness for the case with $h/h_c = 0.8$ and $\nu = 1.4$ m/s. As shown in Fig. 6a, upon collision with the cubic object, the granular flow undergoes a pronounced symmetric deflection. The object acts as an obstacle to the bulk of the granular flow. As a result, a substantial portion of the particles is forced to bypass the object from both lateral sides and continues to propagate downstream, while a smaller fraction accumulates and climbs along the object. A prominent bow-shock-like wave forms upstream of the object due to the sudden deceleration of

the incoming supercritical flow. This bow shock indicates intense momentum transfer and energy dissipation at the flow-object interface. As the flow expands around the object and moves downstream, the bow shock gradually transitions into a fan-shaped shock wave due to progressive energy dissipation and lateral spreading. On the leeward side of the cube, flow separation gives rise to a low-density wake region, or granular vacuum, characterised by markedly reduced particle concentration. After impact ceases, the cube is visibly displaced from its initial position, indicating that even under shallow-flow conditions the granular flow possesses sufficient momentum to mobilize and transport the object.

To quantify the flow disturbance induced by the cube, Fig. 6b presents the vertical profiles of particle streamwise velocity in two representative regions marked in Fig. 6a. Region A is located immediately upstream of the cube and adjacent to its front face, and therefore represents the strongly object-perturbed region. Region B is located in the middle of the lateral bypassing flow beside the cube and is aligned with region A in the streamwise direction, representing the weakly object-perturbed region. Both sampling regions have identical dimensions, with a streamwise length of $4.0D$ and a transverse width equal to the cube width h_c . To obtain the velocity profiles, each sampling region is divided into a series of vertical bins, and the streamwise velocity component of all particles within each bin is averaged. At the initial impact moment ($t = 0.004$ s), the velocity in region A is substantially lower than that in region B, demonstrating a strong local deceleration of particles. Simultaneously, the velocity of the cubic object (grey dashed line) increases rapidly from zero to approximately 0.47 m/s due to intensive frontal impact, indicating efficient transfer of momentum from the flow to the object. As the impact develops, the velocity difference between regions A and B gradually diminishes due to continuous

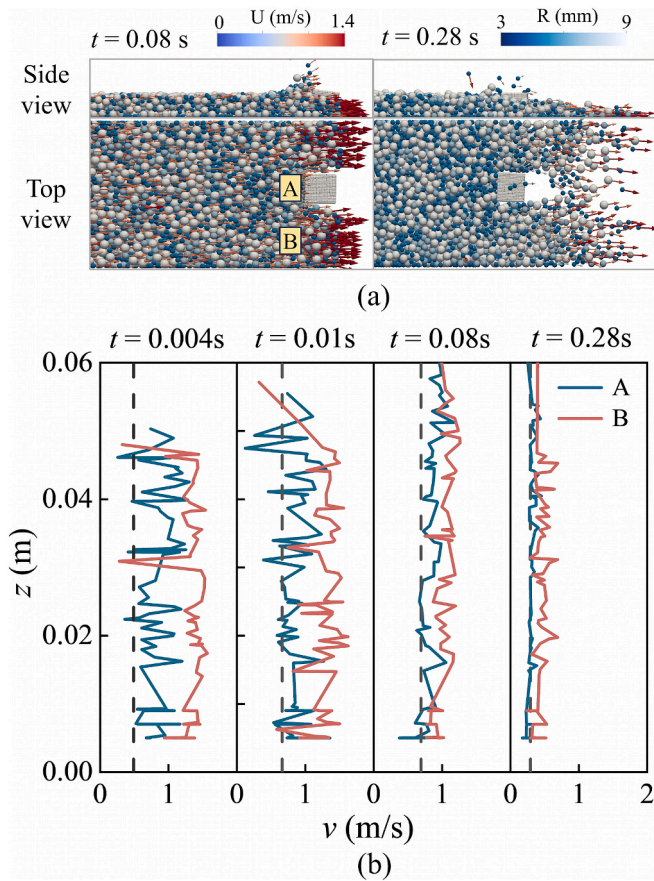


Fig. 6. (a) Flow kinematics and velocity field evolution of granular flow impacting the cubic object ($h/h_c = 0.8$ and $v = 1.4$ m/s); (b) Vertical profiles of the streamwise velocity in the strongly object-perturbed region A and the weakly object-perturbed region B. Both regions have identical dimensions of $4.0D \times h_c$. Grey dashed line denotes the streamwise translational velocity of the cube.

momentum redistribution and attenuation of the local disturbance. More importantly, the cube velocity gradually approaches the particle velocity in the surrounding flow, particularly that in the weakly disturbed region B. This observation supports the key assumption made in the analytical model (Section 2), where the object and the surrounding granular material tend to move synchronously with minimal relative velocity after an initial acceleration of the object to the velocity comparable to the bulk granular flow.

For comparison, Fig. 7 illustrates the dynamic impact and burial process of the cubic object under a deeper granular flow condition ($h/h_c = 3$) with the prescribed initial velocity $v = 3$ m/s. To better trace both the object motion and the associated flow structure, both side-view and plan-view cross-sections are presented in Fig. 7a. Similar to the shallow-flow case (Fig. 6), a bow-shock-like compression region still forms upstream of the object. However, due to the greater flow depth, the incoming particles rapidly overtop the object. A large volume of granular material climbs over the top and sides of the cube, leading to progressive burial of the object. Unlike the previous case where most particles were deflected laterally, here the flow engulfs the object from above, resulting in a more pronounced run-up and eventual coverage of the object by the granular mass. In the early stage, a low-density wake region is also generated behind the object ($t = 0.12$ s). However, unlike the shallow-flow case, this wake region does not persist for long. As more particles continue to pass over the top of the object, the wake region is rapidly filled, and the local particle concentration behind the object increases significantly ($t = 0.24$ s). Therefore, the dominant mechanism in Fig. 7a is not lateral bypassing but rather overtopping,

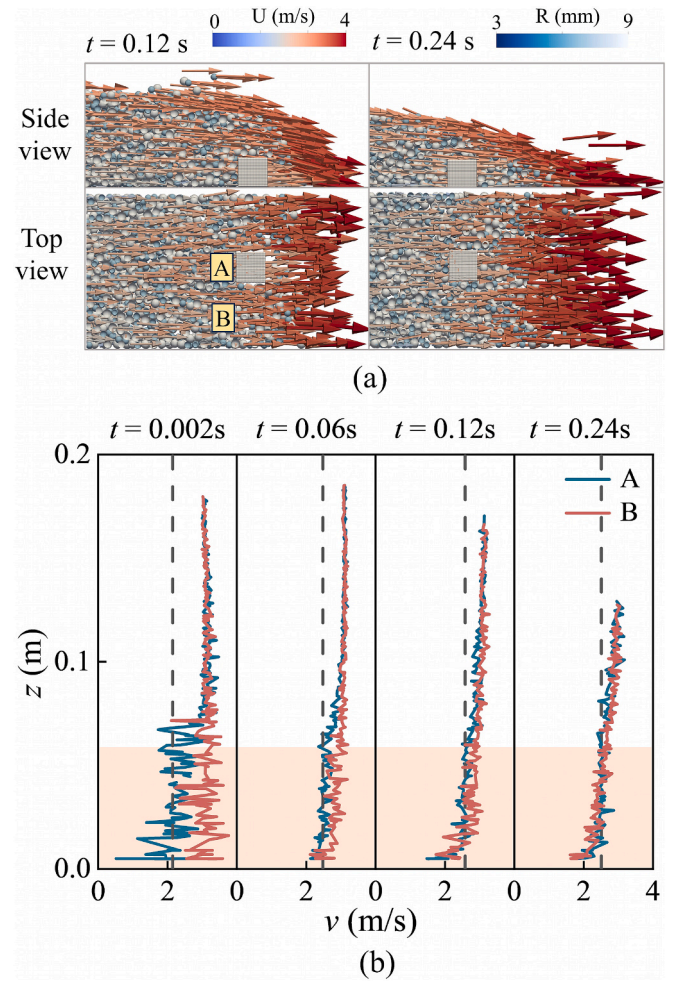


Fig. 7. (a) Flow kinematics and velocity field evolution of granular flow impacting the cubic object ($h/h_c = 3.0$ and $v = 3.0$ m/s); (b) Vertical profiles of the streamwise velocity in the strongly object-perturbed region A and the weakly object-perturbed region B. Both regions have identical dimensions of $4.0D \times h_c$. The pink shaded region indicates the vertical extent of the cube height. Grey dashed line denotes the streamwise translational velocity of the cube.

engulfment, and eventual burial of the cube by the deep granular flow.

The corresponding velocity profiles in Fig. 7b further demonstrate how the object redistributes the surrounding flow momentum. The same regions A and B and the same vertical-bin averaging procedure are used in Fig. 7b. The pink shaded region indicates the vertical extent of the cube height. The streamwise velocity in the strongly perturbed region A remains lower than that in region B ($t = 0.002$ s and 0.06 s), confirming the decelerating effect of the object on nearby particles. Nevertheless, this disturbance is relatively limited when compared with the full flow thickness, which is mainly concentrated within a zone comparable to the object height. Above this zone, the upper part of the flow is able to pass over the obstacle with comparatively less disturbance. As the flow overtops and buries the object, the velocity of the cube becomes increasingly coupled with that of the surrounding granular material. This synchronization is reflected by the gradual convergence of the cube velocity and the particle velocities measured in regions A and B ($t = 0.12$ s and 0.24 s), which again supports the assumptions made in the analytical model (Section 2).

The comparison between Fig. 6 and Fig. 7 highlights two fundamentally different interaction mechanisms between granular flow and movable object. In the shallow-flow case ($h/h_c \leq 1$, Fig. 6), the object acts as an obstacle. The dynamic response is therefore dominated by

frontal impact, strong lateral deflection, and upstream accumulation. The object is mainly transported by the horizontally deflected flow, while burial remains limited. In contrast, in the deeper-flow case ($h/h_c > 1$, Fig. 7), the incoming flow is sufficient to overtop the object. The interaction is then governed primarily by vertical engulfment and burial rather than by lateral bypassing alone. The object becomes increasingly embedded within the moving granular mass.

These observations indicate that the transition from non-burial to burial is a key threshold in flow-object interaction, because it changes the dominant momentum-transfer pathway from lateral deflection around an exposed object to flow engulfment and kinematic coupling within a submerged state. It should be emphasized, however, that $h/h_c = 1$ is used here as an indicator rather than a universally sharp physical boundary. In the shallow-flow case with $h/h_c = 0.8$ shown in Fig. 6, a small number of particles climb along the upstream face of the cube and overtop its upper surface. Nevertheless, this overtopping is transient and the overlying particle layer remains shallow. The cube remains largely exposed, and its motion is dominated by frontal impact and lateral flow deflection rather than sustained engulfment. Therefore, this case is classified as predominantly unburied. In contrast, for $h/h_c > 1$, the incoming flow persistently overtops and progressively engulfs the cube, leading to sustained burial and stronger kinematic coupling with the surrounding granular mass. Thus, the buried/unburied classification adopted here should be interpreted as a simplified regime classification based on the dominant interaction mode observed in the DEM simulations.

4.2. Impact force

A quantitative analysis of the impact force acting on the cubic object is essential, as it directly governs the object's response, including its acceleration, displacement, and burial. To identify the key physical controls on the impact force, it is more effective to employ dimensionless parameters that encapsulate the combined effects of the principal system variables. In this study, the two primary dimensionless flow parameters are the Froude number, Fr , and the dimensionless flow discharge, q^* . These dimensionless parameters collectively incorporate the effects of the incoming flow velocity v , flow thickness h , grain size D , and gravity g . Evidently, the impact force F is also governed by these parameters and can be expressed in the following general form:

$$F = f_F(v, h, D, g, \text{other parameters}) \quad (19)$$

where the “other parameters” include bulk flow density, object density, object height, object width, density ratio, and relevant frictional parameters. These parameters are physically important for a finite-width movable object because they affect the projected area, object inertia, and momentum-transfer efficiency. They are not considered unimportant; rather, they are kept fixed in the present simulations. The object is always represented as an idealized cube with fixed geometry and density, and the material and contact parameters are also unchanged. Therefore, these object-related and material parameters enter the present dimensional analysis as constants. Following the strategy of seeking the shortest possible list of governing parameters (Parker et al., 2007), we focus on the principal varying flow parameters, namely v , h , D , and g , and examine whether the normalized impact force can be correlated with either Fr or q^* . Accordingly, the force is normalized as $F/(\rho v^2 h^2)$, which is obtained from dimensional analysis and represents a dimensionless impact force. The resulting collapse should be interpreted as valid for the fixed cubic object geometry and density ratio considered in this study.

In Fig. 8a, $F/(\rho v^2 h^2)$ is plotted against Fr . In this study, the impact force F is used as the time-averaged streamwise contact force acting on the cubic object during impact. For a given relative flow thickness h/h_c , $F/(\rho v^2 h^2)$ decreases with increasing Fr . On a log–log scale, an approximately linear, monotonic relationship is observed, although the slope varies

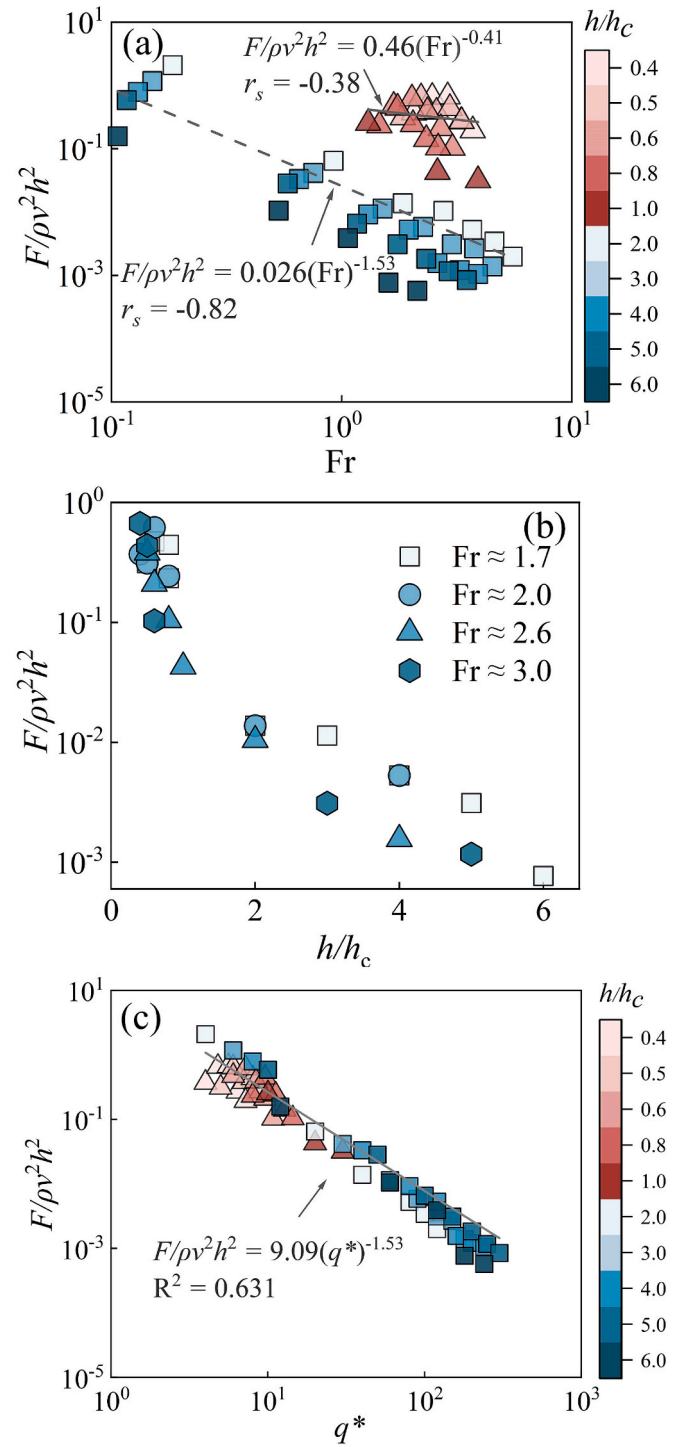


Fig. 8. (a) Dimensionless impact force $F/(\rho v^2 h^2)$ as a function of the Froude number Fr . r_s is the Spearman's correlation coefficient. (b) Dependence of dimensionless impact force $F/(\rho v^2 h^2)$ on h/h_c for cases within comparable Fr ranges. (c) Dimensionless impact force $F/(\rho v^2 h^2)$ versus the dimensionless flow discharge q^* . R^2 is the coefficient of determination. The triangular symbols denote shallow-flow cases ($h/h_c \leq 1$), which are classified as predominantly unburied, whereas the square symbols denote deep-flow cases ($h/h_c > 1$), in which sustained overtopping and engulfment lead to burial of the object by the granular mass.

with h/h_c . As shown in Fig. 8a, the data do not collapse onto a single curve; instead, they separate into two distinct branches. This separation is consistent with the two impact mechanisms identified in Figs. 6 and 7, namely unburied regime ($h/h_c \leq 1$) and buried regime ($h/h_c > 1$). This indicates that Fr , which primarily characterizes the ratio of flow inertia to gravity, is insufficient to uniquely describe the normalized impact force when the relative flow thickness and burial state vary. In particular, the effective portion of the flow that directly interacts with the cube changes between the two regimes: shallow flows are mainly arrested and deflected by the exposed cube, whereas deeper flows can overtop and engulf the cube.

To further clarify the role of the relative flow thickness, Fig. 8b plots $F/\rho v^2 h^2$ against h/h_c for cases within comparable Fr ranges. The normalized impact force generally decreases with increasing h/h_c . This trend indicates that thinner flows transfer momentum to the cube more efficiently, because a larger fraction of the incoming flow depth is directly intercepted and arrested by the exposed object. In contrast, as h/h_c increases, the cube occupies a smaller fraction of the total flow depth. A substantial portion of the granular material can overtop the cube or bypass it laterally, and therefore does not contribute uniformly to the direct impact force. Consequently, the impact force does not scale simply with the flow thickness. This interpretation is consistent with the velocity profiles in Fig. 7b, where the disturbance induced by the cube is mainly concentrated within a zone comparable to the cube height, while the upper part of the flow passes over the object with comparatively less disturbance.

To quantify this regime dependence shown in Fig. 8a, power-law relationships are fitted separately to the two groups, and the corresponding Spearman correlation coefficients, r_s , are calculated. For the unburied cases, a relatively weak negative correlation is obtained: $F/\rho v^2 h^2 = 0.46(Fr)^{-0.41}$ with $r_s = -0.38$. In contrast, for buried cases, $F/\rho v^2 h^2$ shows a much stronger dependence on Fr : $F/\rho v^2 h^2 = 0.026(Fr)^{-1.53}$ with $r_s = -0.82$. These results confirm that the apparent Fr -dependence of the normalized impact force is strongly regime-dependent, and that Fr alone cannot provide a unified scaling across both buried and unburied conditions.

Fig. 8c shows the linkage between $F/\rho v^2 h^2$ and q^* . In this representation, the data collapse onto a single, well-defined trend that can be described by the power-law relation:

$$F/\rho v^2 h^2 = m(q^*)^n \quad (20)$$

where the fitting parameters m and n are 9.09 and -1.53 , respectively, in this study. This behaviour suggests that the impact force is governed more directly by the incoming granular discharge. Such a result is physically reasonable, because the impact force originates from momentum transfer from the incoming grains to the object. Therefore, the force depends not only on the bulk flow velocity, but also on the particle-scale discharge. Compared with Fr , which mainly characterizes the bulk inertial state of the flow relative to gravity, q^* incorporates both the incoming velocity and the amount of moving granular material supplied to the object per unit time. It should be noted that Eq. (20) represents a constrained scaling relationship of the dimensionless impact force $F/\rho v^2 h^2$, derived within the adopted dimensional framework and under the specific conditions considered here. Alternative normalizations, such as projected area, may lead to different fitted coefficients and collapse quality.

4.3. Travelling distance

To further quantify the mobility of the cubic object under granular-flow impact, the travelling distance s is analyzed in this section. In order to examine the dependence of s on v and h systematically, we perform a standard dimensional analysis by writing s as an unknown function f :

$$s = f(v, h, D, g, \rho_c, \rho, h_c) \quad (21)$$

Buckingham's π theorem enables us to rewrite this general but uninformative equation in equivalent dimensionless forms with fewer independent arguments. We choose to normalize the travelling distance using h and D , respectively, which yields:

$$\frac{s}{h} = \Phi_1\left(\frac{v}{\sqrt{gh}}, \frac{\rho}{\rho_c}, \frac{h}{h_c}, \frac{h}{D}\right) \quad (22a)$$

and

$$\frac{s}{D} = \Phi_2\left(\frac{vh}{\sqrt{gD^3}}, \frac{\rho}{\rho_c}, \frac{h_c}{D}\right) \quad (22b)$$

where $Fr = v/\sqrt{gh}$ is the Froude number and $q^* = vh/\sqrt{gD^3}$ is the dimensionless flow discharge. ρ/ρ_c is the density ratio and h/h_c is the relative flow thickness, and h_c/D is the ratio of object height to grain size. Since $h/D = (h/h_c)(h_c/D)$, h/D is not an independent parameter. In this study, only v and h are varied, while the material (ρ/ρ_c) and geometric parameters (h_c/D) are fixed. We therefore focus primarily on the flow-related dimensionless groups Fr and q^* , while using h/h_c as an

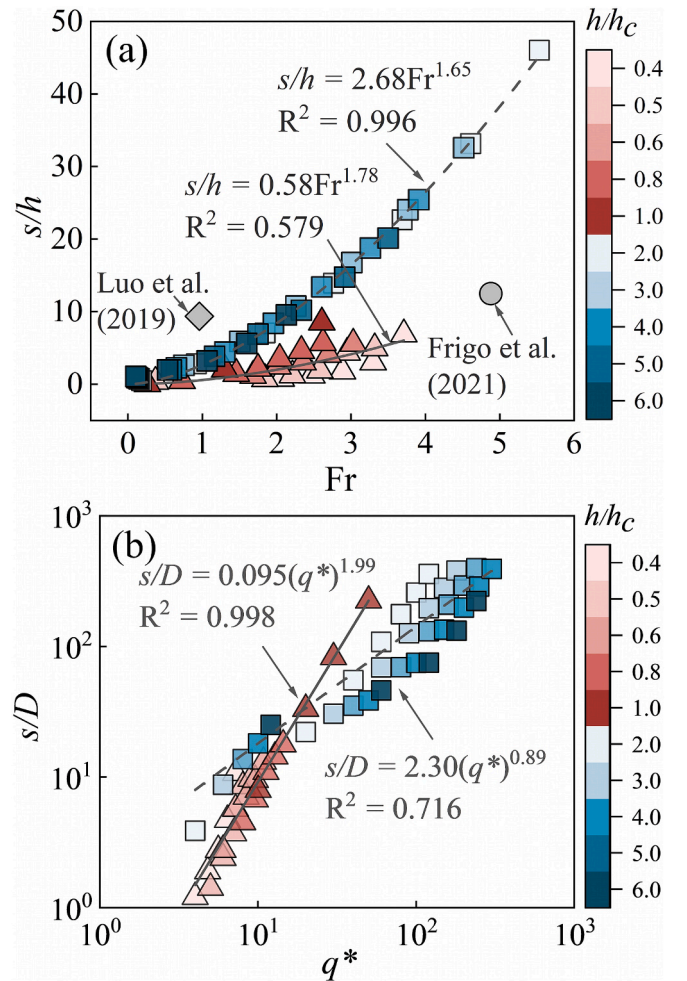


Fig. 9. (a) Dimensionless travelling distance normalized by flow thickness s/h as a function of the Froude number Fr . (b) Dimensionless travelling distance normalized by grain size s/D as a function of the dimensionless flow discharge q^* . R^2 is the coefficient of determination. The triangular symbols denote shallow-flow cases ($h/h_c \leq 1$), which are classified as predominantly unburied, whereas the square symbols denote deep-flow cases ($h/h_c > 1$), in which sustained overtopping and engulfment lead to burial of the object by the granular mass. Natural wood-snow avalanche (Frigo et al., 2021) and debris landslide (Luo et al., 2019) cases are also included.

indicator to distinguish between predominantly unburied and buried regimes (Fig. 9). Based on the dimensional analysis in Eqs. (22a) and (22b), two alternative normalizations of the travelling distance are examined. The first uses the flow thickness h as the characteristic length scale, leading to s/h as a function of Fr , because $Fr = v/\sqrt{gh}$ is defined using h . The second uses the grain-size scale D , leading to s/D as a function of q^* , because $q^* = vh/\sqrt{gD^3}$ represents the dimensionless granular discharge at the particle scale.

Fig. 9a shows the relationship between s/h and Fr . For both unburied (triangle symbols) and buried (square symbols) cases, s/h increases monotonically with increasing Fr . However, the two regimes exhibit markedly different collapse behaviour. For the buried cases, the data collapse well onto a single trend, indicating that once the object is buried, its displacement is primarily governed by the inertial state of the surrounding granular flow. In this regime, the flow thickness h becomes the relevant length scale, and the displacement can therefore be characterised effectively by Fr . In contrast, the unburied cases show a poorer collapse and exhibit a much stronger dependence on the relative flow thickness h/h_c . This indicates that, when the cube is not buried, s/h is not controlled by Fr alone but is still highly sensitive to the degree of submergence.

Fig. 9b replots the displacement data in terms of s/D versus q^* . Similarly, s/D increases monotonically with q^* , but the regime dependence is opposite. The unburied cases collapse remarkably well, indicating that the displacement of an unburied object is primarily controlled by the incoming granular discharge. Physically, this suggests that, before burial occurs, the cube mobility is governed mainly by the transport capacity of the impacting granular flow, which is more naturally captured by q^* than by Fr . The buried cases exhibit a stronger dependence on h/h_c , implying that q^* is no longer the dominant descriptor once the object is embedded in the granular mass.

Fig. 9 indicates that displacement is controlled by different mechanisms in the unburied and buried regimes. When the cube remains unburied, its motion is governed mainly by the incoming granular discharge, so that q^* is the more relevant controlling parameter. Once the cube becomes buried, however, its movement is increasingly constrained by the surrounding granular material, and the flow thickness h becomes increasingly important. Accordingly, Fr becomes the more appropriate descriptor. The corresponding empirical expressions are

$$s/D = a_1(q^*)^{b_1}, \quad h/h_c \leq 1 \quad (23a)$$

and

$$s/h = a_2(Fr)^{b_2}, \quad h/h_c > 1 \quad (23b)$$

where a_1 , b_1 , a_2 and b_2 are fitting parameters, with values of $a_1 = 0.095$, $b_1 = 1.99$, $a_2 = 2.68$ and $b_2 = 1.65$ here. In addition, data from a natural wood-snow avalanche (Frigo et al., 2021) and a natural debris landslide (Luo et al., 2019) are included in Fig. 9a. In these two burial cases, the reported building displacements are approximately 50 m and 150 m, respectively. The discrepancy between the two natural cases and the numerical simulations likely arises because the field objects are not transported as idealized free-sliding rigid blocks. In natural settings, additional processes such as foundation failure, overturning, structural damage, and heterogeneous basal resistance can strongly modify the displacement behaviour. These processes would alter the momentum-transfer and energy-dissipation mechanisms, but they are not explicitly represented in the present DEM simulations. Therefore, the proposed relationships should be interpreted as applicable primarily to intact rigid cubic objects transported in regimes dominated by impact, burial, and downstream displacement. In addition, real structures may deform or disintegrate upon impact. The present DEM object is treated as a rigid clump, and thus ignores structural deformation, breakage, or progressive mass loss. Such processes would alter the object shape, mass distribution, effective contact area, and energy dissipation. Future work

should therefore consider deformable or breakable objects and examine whether the proposed scaling relationships remain valid when rolling, overturning, twisting, or fragmentation becomes significant.

4.4. Energy transfer

To better understand the dynamic impact and displacement responses of the object, the granular flow-object interaction is examined here from an energy perspective. Fig. 10 shows the temporal evolution of the unit-mass energies of the granular flow-object system, indicating that impact process is fundamentally a process of energy redistribution. E_{kf} , E_{kc} , and ΔE represent the unit-mass kinetic energy of the granular flow, the unit-mass kinetic energy of the object, and the unit-mass reduced energy of granular flows, respectively. Here, the kinetic energy of incoming flow ϵ_{kf} is calculated as follows:

$$\epsilon_{kf} = \frac{1}{2} \sum_{i=1}^n (m_i v_i^2 + I_i w_i^2) \quad (24)$$

where w_i and v_i are the angular and translational velocities of particle i , respectively. m_i is the mass of particle i . I_i is its moment of inertia. n is the total number of particles. E_{kf} is obtained as $E_{kf} = \epsilon_{kf} / \sum_{i=1}^n m_i$. In light of the principle of energy conservation, ΔE of the granular flow at any given time can be calculated as:

$$\Delta E = (\epsilon_0 - \epsilon_p - \epsilon_{kf}) / \sum_{i=1}^n m_i \quad (25)$$

where ϵ_0 is the initial total energy of the granular flow, ϵ_p is the potential energy and is given as $\sum_{i=1}^n m_i g h_i$. h_i is the height of particle i . The kinetic energy of the object ϵ_{kc} is calculated as

$$\epsilon_{kc} = \frac{1}{2} (m_c v_c^2 + I_c w_c^2) \quad (26)$$

where w_c and v_c are the angular and translational velocities of the cube, respectively. I_c is the cube's moment of inertia. E_{kc} is obtained as $E_{kc} = \epsilon_{kc} / m_c$, where m_c is the mass of the cube. Fig. 11 compares the temporal evolution of the translational and rotational kinetic energies of the cube for representative unburied and buried cases. The rotational kinetic energy is much smaller than the translational kinetic energy throughout the transport process, indicating that the cube motion is dominated by downstream translation. In the unburied case, minor rotation occurs, whereas in the buried case, rotation is further suppressed by the confinement of the surrounding granular material. R_E denotes the energy conversion ratio and is defined as the ratio of the energy obtained by the object to the reduced energy of the granular flow, which is given as

$$R_E = \frac{1}{2} (m_c v_c^2 + I_c w_c^2) / (\epsilon_0 - \epsilon_p - \epsilon_{kf}) \quad (27)$$

As the granular flow impinges on the object, E_{kf} decreases continuously until reaching a stable level, as shown in Fig. 10. At the same time, ΔE increases monotonically but at a decreasing rate and eventually approaches a stable value. The reduced energy of granular flow is partitioned into two main components: one part is transferred to the object and drives its motion, while the other is dissipated through grain-scale interactions within the flow and through flow-object interaction. Correspondingly, E_{kc} rapidly increases to the maximum during the initial stage of impact and then gradually decays to zero, as the transferred energy is not sufficient to overcome frictional or collisional dissipation. Likewise, R_E reaches its maximum during the early stage of impact and then decays rapidly. The maximum value of R_E remains below 3.6%, indicating that only a small fraction of the flow energy is converted into object motion, whereas most of the energy is dissipated.

Clear differences are observed between the unburied and buried regimes. Compared to the unburied case ($h/h_c = 0.8$, Fig. 10a-10b), the

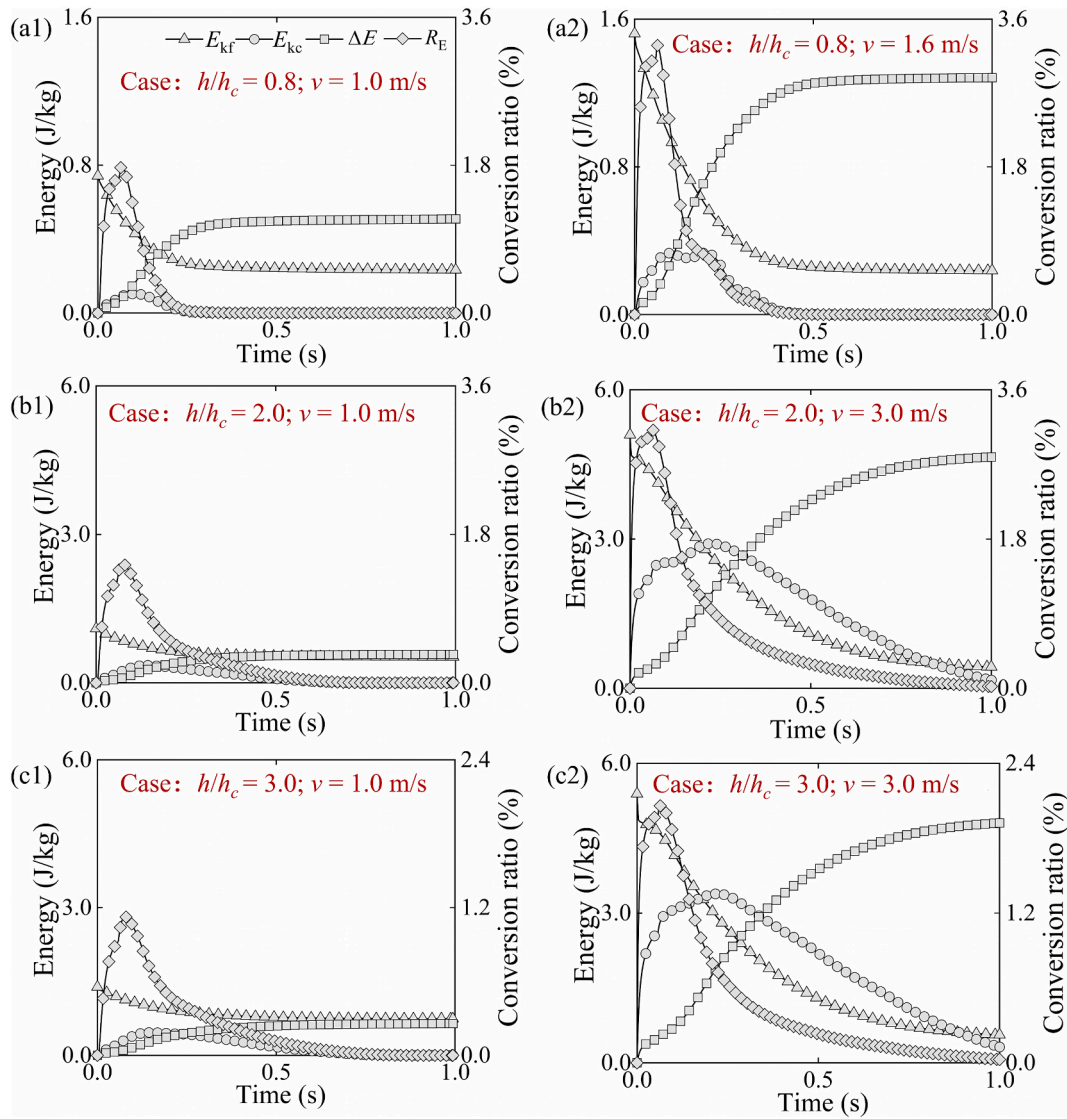


Fig. 10. Evolution of unit-mass energy for different controlling parameters: (a, b): $h/h_c = 0.8$, $v = 1.0$ m/s and 1.6 m/s; (c, d): $h/h_c = 2.0$, $v = 1.0$ m/s and 3.0 m/s; (e, f): $h/h_c = 3.0$, $v = 1.0$ m/s and 3.0 m/s.

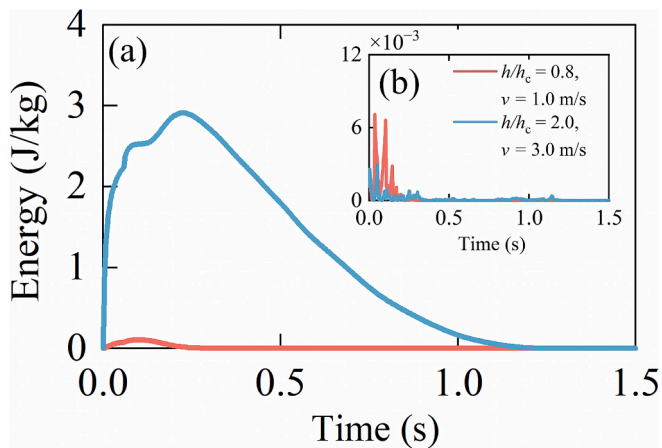


Fig. 11. Temporal evolution of the specific kinetic energy of the cubic object for representative unburied and buried cases: (a) translational kinetic energy and (b) rotational kinetic energy. Red lines denote the unburied case ($h/h_c = 0.8$, $v = 1.0$ m/s) and blue lines denote the buried case ($h/h_c = 2.0$, $v = 3.0$ m/s).

peak of E_{kc} is much higher and its decay is more gradual, and ΔE continues to grow over a longer period and reaches a larger final value for the buried cases ($h/h_c = 2$ and 3, Fig. 10c–10f). These differences are physically attributed to different flow-object interactions. When the object remains unburied, the interaction is dominated by short-duration frontal impact and lateral flow deflection. Once the object becomes buried, the object undergoes sustained interaction with the surrounding granular mass, which prolongs both energy transfer to the object and energy dissipation within the system.

The effects of flow velocity and flow thickness are also evident. For a given h/h_c , increasing v leads to a higher initial E_{kf} , a larger peak E_{kc} , a greater dissipation ΔE and a larger peak R_E . Similarly, for a given v , increasing h/h_c raises the initial E_{kf} , the peak E_{kc} , the dissipation ΔE and the peak R_E . Taking R_E as an example, when h/h_c increases from 0.8 to 2, the maximum value of R_E increases from 1.8% to 2.4% for a constant $v = 1$ m/s. When v increases from 1 m/s to 3 m/s, the maximum value of R_E increases from 1.1% to 2.1% for a constant $h/h_c = 3$. Overall, Fig. 10 provides an energy-based interpretation in the unburied and buried regimes.

5. Discussion

5.1. Estimation of dynamic impact

Evaluating the dynamic impact exerted by granular flow on the object remains a major concern. However, this has been limited by the difficulty in reliably determining dynamic pressure coefficient α . Reported values of α for granular flow span a remarkably wide range from 0.4 to > 18 (Hu et al. 2011), likely due to differences in material composition and natural conditions (Song et al., 2017; Poudyal et al., 2019). A power law correlation between α and Fr has commonly been adopted (Hübl et al., 2009; Scheidl et al., 2013): $\alpha = c \cdot Fr^{-d}$, where c and d are positive constants. The value of α has been explored through both laboratory experiments at different scales and numerical simulations (Hungr et al., 1984; Cui et al., 2015; Fang et al., 2022). On the basis of extensive field and experimental data, Cui et al. (2015) fitted the empirical relation: $\alpha = 5.3Fr^{-1.5}$. Despite these efforts, the practical selection of α is still not supported by rigorous and universally applicable scientific guidance, and it is often determined using subjective judgement and engineering experience. This inevitably introduces additional uncertainty into predictions of object travelling distance. It is also worth noting that previous studies have mainly focused on impacts against fixed structural obstacles, such as rigid barriers (Song et al., 2017; Fang et al., 2022) and circular obstacles (Yu et al., 2026), whereas impacts on movable objects have received much less attention.

Fig. 12 compares the dynamic pressure coefficient α as a function of the Froude number Fr using data from previous studies (Hübl & Holzinger, 2003; Proske et al., 2011; Bugnion et al., 2012; Scheidl et al., 2013; Cui et al., 2015; Li et al., 2020), together with the results of the present study. Overall, α decreases systematically with increasing Fr in this study, indicating a clear negative power-law dependence. This trend is generally consistent with previous experimental and numerical studies, suggesting that the inverse relationship between α and Fr is a robust feature of granular-flow impact. Nevertheless, the α values obtained in the present study are systematically lower than those reported in most previous studies. The combined buried and unburied data are well described by $\alpha \sim Fr^{-1.67}$, which deviates from the canonical exponent of -1.5 reported by Cui et al. (2015). The lower α values in the present study can be attributed to the fact that the cubic object simulated in this study can be regarded as a movable obstacle with finite width. Its motion reduces the relative velocity during impact, while its finite width allows part of the granular flow to bypass the object, thereby reducing the efficiency of momentum transfer. This discrepancy would result in systematic overestimation of the impact force when applying hydrodynamic-based predictions with the values of α from Cui et al.

(2015), which consequently underestimate the predicted travelling distance according to Eq. (14). Therefore, the fitted $\alpha \sim Fr$ relation obtained from the combined buried and unburied data is used in the subsequent theoretical calculations of travelling distance (Eq. (14)).

It should be noted that the $\alpha \sim Fr$ relation obtained in this study is based on an idealized prescribed-inflow setup. In the DEM simulations, all particles are assigned the same initial velocity, resulting in an initially plug-like velocity profile rather than a fully developed sheared profile. Although this treatment enables systematic control of the bulk incoming velocity and effective flow depth at the cube location, it may underrepresent the effects of vertical velocity gradients induced by basal friction. In realistic granular flows, particles near the base may move more slowly than those near the free surface. Such a velocity gradient could alter the distribution of momentum transfer to the object, generate additional torque or rolling, and consequently influence burial and travelling distance. Therefore, the present $\alpha \sim Fr$ relation should be interpreted as an idealized case of granular flow impact under controlled inflow conditions. Future work should examine the sensitivity of object motion to more realistic inflow structures, such as gravity-driven release over a longer upstream distance or prescribed sheared velocity profiles.

5.2. Travelling distance prediction

Fig. 13 assesses the ability of the analytical models developed in Section 2 to predict object travelling distance through comparison with the DEM results. The comparison shows that model performance is strongly regime-dependent. For the buried cases (square symbols, $h/h_c > 1$), the predicted travelling distances show good overall agreement with the DEM results, with most data points clustering closely around the 1:1 line. In contrast, for the unburied cases (triangle symbols, $h/h_c \leq 1$), the predictions exhibit greater scatter and a tendency towards overestimation. Nevertheless, the majority of data points from both regimes still lie within a reasonable range around the 1:1 correspondence line. This indicates that the analytical models are capable of providing a physically meaningful description of how granular-flow impact drives the displacement of a movable object.

It should be noted, however, that Eqs. (4) and (16) (in ‘Section 2’) are attractive in displacement prediction because of their mathematical simplicity and clear physical meaning, but they also have inherent limitations. In particular, they do not account for particle-scale force variations associated with momentum transfer, they neglect the rotation of the object, and they treat the flow velocity v and flow depth h as constants, thereby neglecting possible spatial and temporal variations in the flow. These simplifications are likely responsible for part of the deviation between the analytical predictions and the DEM results.

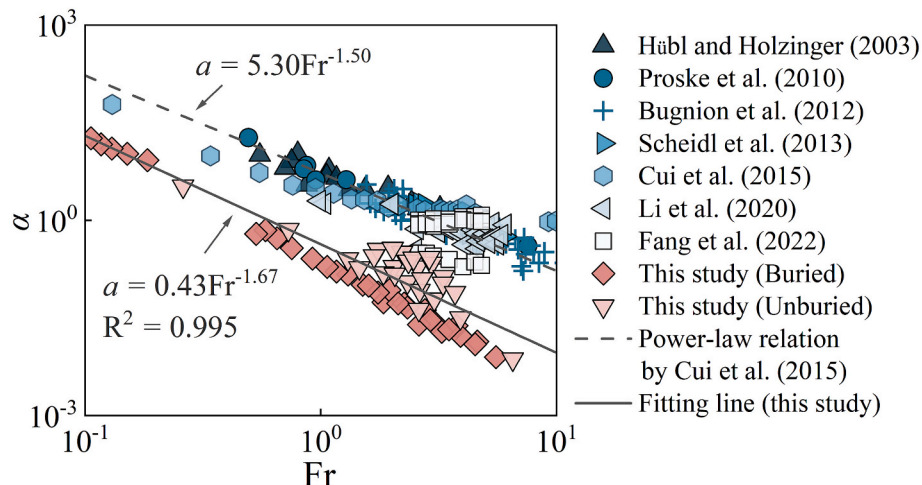


Fig. 12. Relationship between the dimensionless impact force coefficient α and the Froude number Fr .

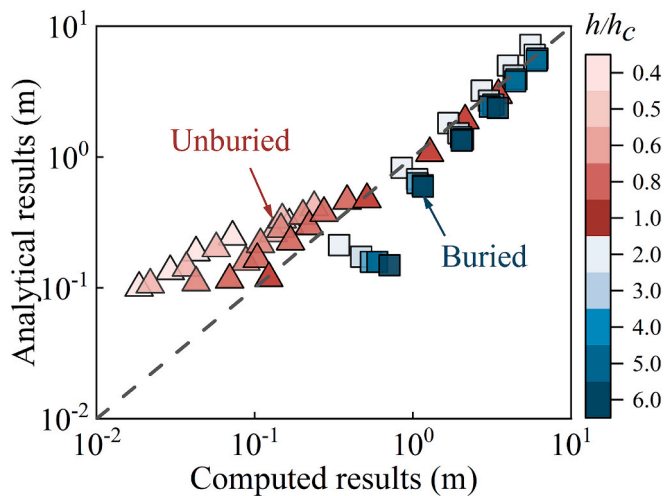


Fig. 13. Comparison of analytical and computed travelling distance. Grey dashed line represents a 1:1 match between theoretical and computed results. The triangular symbols denote unburied regime, whereas the square symbols denote buried regime.

Despite these limitations, the analytical models remain valuable. Their purpose is not to reproduce the full complexity of multi-scale interactions, but to provide a tractable predictive tool for estimating travelling distance using a limited number of physically interpretable parameters. Importantly, the model captures the correct order of magnitude of the displacement and successfully distinguishes the contrasting trends between the buried and unburied regimes. This indicates that the dominant mechanisms governing object motion are reasonably represented in the present formulation. Therefore, the analytical models can be regarded as a useful framework for travelling distance estimation.

For practical use of the analytical solutions developed in Section 2, the burial regime should first be determined using the ratio h/h_c . If $h/h_c \leq 1$, the object is treated as unburied and Eq. (4) is used. If $h/h_c > 1$, the object is treated as buried and Eq. (16) is used. The required input parameters include the object height h_c , object density ρ_c , bulk flow density ρ , flow depth h , flow velocity v , flow length L , and basal friction parameters μ_c . For the buried-object model, additional parameters, including the dynamic pressure coefficient α , earth pressure coefficient k , and flow-object friction coefficient μ_1 are also required. In real events, the flow depth may be estimated by comparing pre- and post-event channel-bed topography using drone-based photogrammetry, flow velocity may be inferred from recorded video analysis (De Haas et al., 2020). Because these quantities are uncertain under real post-event conditions, the predicted travelling distance should be interpreted as a range rather than a single deterministic value. For the unburied model, Eq. (4) indicates that the predicted distance is particularly sensitive to the flow velocity, because s scales with v^2 when other parameters are fixed. Thus, uncertainty in v is amplified in the prediction of the travelling distance. For the buried model, Eq. (16) is more nonlinear because the dynamic impact, basal resistance, lateral pressure, and surrounding granular confinement are coupled. In practical applications, the formulas can therefore be used with plausible upper and lower bounds of the input parameters to estimate a possible downstream search-zone range rather than a unique final location.

5.3. Implications

The results of this study have practical implications for post-disaster search and rescue. A major challenge during rescue operations is that movable objects, such as houses, vehicles, and containers, may be displaced from their original positions and eventually come to rest at unknown downstream locations, where they can also be partially or fully

buried. In this context, the regime-dependent transport distance identified in this study provides a physically based explanation of the mechanisms governing object displacement and burial, and may therefore help improve the interpretation of post-event object locations and support more targeted search and rescue efforts.

An important remaining question, however, is whether the physical interpretation embodied in Eq. (23) can be extended from laboratory-scale conditions to field-scale granular flows. To explore this possibility, we conducted an additional large-scale flume test of a flow-like landslide impacting a 15-cm cubic object at the Kadoorie Centre of the University of Hong Kong. It should be emphasized that this comparison is intended only as a preliminary large-scale check, rather than a rigorous validation of field-scale applicability. The flume has a total length of 28 m, a width of 2 m, and a depth of 1 m. More details of the large-scale model facility can be found in Ng et al. (2024). The incoming flow material consists of 35% gravel with a representative size of 20 mm, 62.5% sand with a representative size of 0.6 mm, and 2.5% clay with particle size smaller than 2 μm . The cubic object had a side length of $h_c = 0.15$ m and a density of 2074 kg/m^3 , which is placed on the horizontal concrete floor at the downstream end of the flume. The impact process is recorded by video camera, which is available at 10.5281/zenodo.20437296.

The video observations show that the cubic object remains unburied during transport. No obvious overturning is observed. However, because the object is partly obscured by the incoming flow material, minor rocking or rotation could not be quantified accurately from the available video images. By analyzing the recorded images, the flow velocity and flow depth immediately before impact at the object location are estimated to be 5.4 m/s and 0.048 m, respectively. The white grids with the dimension of 1 $\times$ 1 m are marked to measure the travelling distance, which is determined as 6.6 m. Following Ng et al. (2024), the representative grain size is taken as 0.01 m. Substituting these values into Eq. (23a) yields a predicted transport distance of 6.2 m, which differs from the measured value by about 6%.

This agreement should be interpreted with caution. The corresponding Froude number of the field-scale case is $\text{Fr} \approx 7.9$, indicating a highly inertial flow condition that is relevant to rapid natural granular flows. The large-scale flume case is not strictly dynamically similar to the DEM simulations because of different material grading, basal roughness, object density, flow rheology, boundary conditions, and scale. Nevertheless, both the DEM simulations and the large-scale test involve rapid flow-object interaction in which grain contacts, basal resistance, and frictional momentum transfer play important roles. Therefore, the comparison is interpreted at the level of dominant flow-object mechanics rather than exact dynamic similarity.

The video-based estimates of flow depth, flow velocity, and travelling distance are subject to uncertainties associated with image resolution, frame rate, perspective correction, grid resolution, and manual interpretation of the flow front and object position. Despite these limitations, the large-scale flume case provides encouraging preliminary support for the interpretation that, in the unburied regime, the transport distance is primarily governed by the dimensionless flow discharge. This result suggests that the physical interpretation proposed here may have applicability beyond small-scale numerical tests and may be useful for interpreting object transport in larger-scale granular flows, although additional large-scale or field-scale cases are required for broader assessment.

Nevertheless, the transferability of the proposed scaling relationships should be interpreted with caution. First, the large-scale comparison is based on only one unburied case, and additional field-scale or large-scale experimental data are required to further validate the regime-dependent relationships. Second, the DEM simulations are conducted using a fixed weakly polydisperse tridisperse material. This simplified particle-size distribution does not explicitly account for the effects of broader grain-size distributions. In natural granular flows, power-law or Gaussian-like distributions may alter packing density, flow

mobility, and size segregation, and thus local momentum transfer to the object. These effects may further influence the burial process and travelling distance of the cube, especially when coarse particles become preferentially concentrated near the flow front or free surface. Therefore, the present scaling relationships should be interpreted as being applicable to the controlled granular material considered in this study. Although the dimensionless framework identifies the main physical dependencies among flow inertia, granular discharge, burial state, impact response, and travelling distance, it does not imply that the fitted coefficients are invariant for all grain size distributions. Future work should systematically examine how broader grain size distributions affect flow mobility, segregation, object burial, travelling distance, and the resulting scaling relationships. In addition, the possible effects of interstitial fluid and unsaturated grain behaviour are not explicitly represented in the present dry DEM framework. These effects may modify effective stress, grain-contact friction, and bulk rheology (Iverson, 1997), and may require future continuum or hybrid modelling approaches (e.g., Li et al., 2020; Cui et al., 2026).

6. Conclusions

This study investigated the impact, burial, and transport of a movable cubic object subjected to granular flows using three-dimensional DEM simulations and analytical modeling. The main conclusions are as follows:

1. The burial state controls two distinct flow-object interaction regimes. When $h/h_c \leq 1$, the object remains predominantly unburied and is displaced mainly by frontal impact. When $h/h_c > 1$, sustained overtopping and engulfment by the incoming granular material lead to burial of the object, resulting in prolonged flow-object coupling and a fundamentally different transport process.
2. The normalized impact force is better described by the dimensionless flow discharge q^* than by the Froude number Fr . Although $F/\rho v^2 h^2$ decreases with increasing Fr , the data remain sensitive to submergence state. By contrast, the impact force data from both buried and unburied cases collapse well as a function of q^* : $F/\rho v^2 h^2 = 9.09(q^*)^{-1.53}$, indicating that impact force is governed primarily by incoming granular discharge and momentum supply.
3. Travelling distance exhibits clear regime dependence. In the unburied regime, it is mainly controlled by granular discharge and scales with q^* : $s/D = 0.095(q^*)^{1.99}$. In the buried regime, flow thickness becomes the relevant displacement scale, and travelling distance is better described by Fr : $s/h = 2.68(Fr)^{1.65}$. This reflects a transition from transport of an exposed object to transport of an embedded object.
4. Analytical models are developed for both regimes based on their dominant transport mechanisms. The models show good agreement with the DEM results and provide a physically interpretable framework for estimating displaced-object locations.

Overall, this study improves the mechanistic understanding of how burial state controls granular-flow-driven transport of a movable cubic object. The proposed regime-dependent interpretation and analytical models provide a physically interpretable framework for estimating displaced-object locations under the idealized conditions considered here. However, the present study is limited to a single cubic object geometry, one object density, one basal condition, and a simplified granular material. Therefore, the results should not be directly generalized to natural buildings, vehicles, containers, or irregular debris. Future work should extend the analysis to different object shapes, density ratios, basal conditions, material gradings, rotation/overturning behaviour, and larger-scale or field-scale conditions.

CRediT authorship contribution statement

Jun Fang: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Jinrun Wang:** Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Nuo Chen:** Writing – original draft, Investigation, Data curation. **Clarence Edward Choi:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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